

CFD 2025-26 – Lab 10

Exercise 1

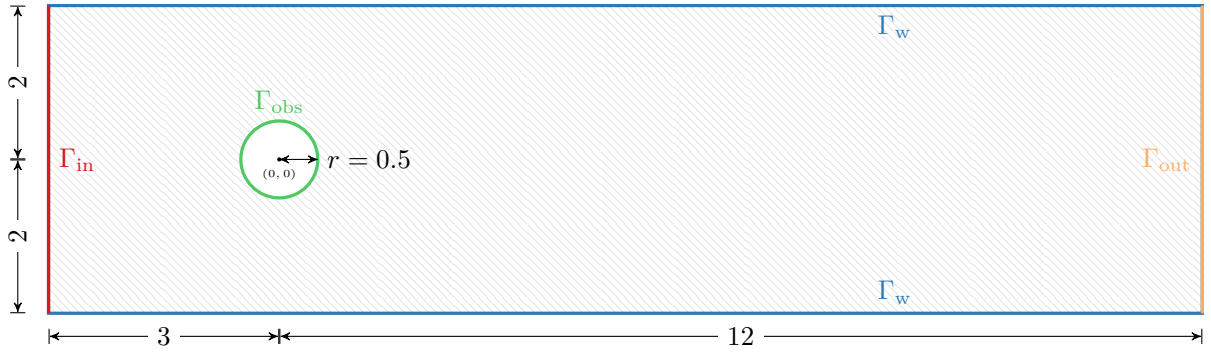
Let us consider the Navier-Stokes problem defined in the domain Ω as in the Figure below

$$\left\{ \begin{array}{ll} \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \frac{1}{\text{Re}} \text{div} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) + \nabla p = \mathbf{0}, & \text{in } \Omega, t \in (0, T), \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, t \in (0, T), \\ \mathbf{u} = \frac{3}{8}(2-y)(2+y)\mathbf{i}, & \text{on } \Gamma_{\text{in}} = \{x = -3, -2 \leq y \leq 2\}, t \in (0, T), \\ \left[\frac{1}{\text{Re}} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) - pI \right] \mathbf{n} = h_{\text{out}} \mathbf{n}, & \text{on } \Gamma_{\text{out}} = \{x = 12, -2 \leq y \leq 2\}, t \in (0, T), \\ \mathbf{u} = \mathbf{0}, & \text{on } \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}}), t \in (0, T), \\ \mathbf{u}|_{t=0} = \mathbf{u}_{\text{Stokes}}, & \text{in } \Omega, t = 0, \end{array} \right.$$

where the Reynolds number Re is equal to 200, boundary conditions are given by:

- no-slip conditions on the lateral boundary of the pipe and on the boundary of the obstacle;
- a parabolic velocity profile on the inlet boundary Γ_{in} ;
- non-homogeneous Neumann boundary conditions at the outlet boundary Γ_{out} , with an outward-directed normal stress $h_{\text{out}} = 0.5$;

and $\mathbf{u}_{\text{Stokes}}$ is the solution of the corresponding steady Stokes problem.



- 1 Discretize the problem using the stable pair of spaces $\mathbb{P}_b^1/\mathbb{P}^1$ and solve the problem using the incremental Chorin-Temam projection scheme coupled with the backward Euler semi-implicit scheme.

Given the solution $\mathbf{u}^n, p^n$ at a certain time instant t^n , the scheme to compute $\mathbf{u}^{n+1}, p^{n+1}$ is described by the following steps:

1) Solve the predictor step to obtain a temporary velocity $\tilde{\mathbf{u}}$:

$$\begin{cases} \frac{\tilde{\mathbf{u}} - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla) \tilde{\mathbf{u}} - \frac{1}{\text{Re}} \text{div}(\nabla \tilde{\mathbf{u}} + \nabla \tilde{\mathbf{u}}^T) = -\nabla p^n, & \text{in } \Omega, \\ \tilde{\mathbf{u}} = \frac{3}{8}(2-y)(2+y)\mathbf{i}, & \text{on } \Gamma_{\text{in}}, \\ \frac{1}{\text{Re}}(\nabla \tilde{\mathbf{u}} + \nabla \tilde{\mathbf{u}}^T)\mathbf{n} = h_{\text{out}}\mathbf{n}, & \text{on } \Gamma_{\text{out}}, \\ \tilde{\mathbf{u}} = \mathbf{0}, & \text{on } \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}}). \end{cases}$$

2) Solve the Poisson problem for the pressure increment δp :

$$\begin{cases} -\Delta \delta p = -\frac{1}{\Delta t} \text{div} \tilde{\mathbf{u}}, & \text{in } \Omega, \\ \delta p = 0, & \text{on } \Gamma_{\text{out}}, \\ \nabla \delta p \cdot \mathbf{n} = 0, & \text{on } \partial\Omega \setminus \Gamma_{\text{out}}. \end{cases}$$

3) Projection step

$$\mathbf{u}^{n+1} = \tilde{\mathbf{u}} - \Delta t \nabla \delta p.$$

4) Pressure update

$$p^{n+1} = p^n + \delta p.$$

- 2 Compute the drag and lift forces exerted by the fluid on the obstacle at the final time $T = 50$, defined as follows:

$$F_D = \int_{\Gamma_{\text{obs}}} \left[p\mathbf{n} - \frac{1}{\text{Re}}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)\mathbf{n} \right] \cdot \mathbf{i} \, ds, \quad F_L = \int_{\Gamma_{\text{obs}}} \left[p\mathbf{n} - \frac{1}{\text{Re}}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)\mathbf{n} \right] \cdot \mathbf{j} \, ds,$$

where Γ_{obs} is the boundary of the obstacle.

NB: Use `FacetNormal(mesh)` to access the normal vector $\mathbf{n}$ in Firedrake.

- 3 Consider now free-slip conditions on the wall boundaries

$$\Gamma_w = \{-3 \leq x \leq 12, y = -2, 2\} = \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}} \cup \Gamma_{\text{obs}}).$$

Rewrite the steps of the projection scheme in this case, modify the implementation and run the simulation. What can you observe?